

# Projection of the Effect of Rising Temperature Caused by Climate Change on Sea Level Rise on Long Island

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## Abstract

The consequences of climate change are far reaching. In the Arctic, the area covered by sea ice has shrunk significantly. The rising sea levels due to the excessive melting of polar ice have led to flooding in some areas, causing disease to spread and damage to ecosystems. Understanding the impact of sea level rise due to climate change is critical for coastal regions, particularly in areas like Long Island, New York, where rising seas pose a significant threat to communities and ecosystems. This research focuses on predicting the effects of sea level change along the entire Long Island coastline over the next 100, 150, and 200 years. The methods employed in this research encompassed the construction of a digital elevation model of Long Island using GIS software (Global Mapper v.24.1) to identify affected areas. Models are constructed utilizing data from NYSDEC (New York State Department of Environmental Conservation) projection for sea level rise on Long Island. By analyzing historical and projected climate data, this study identifies the overall area and the specific regions at risk of flooding under various sea level rise scenarios.

Based on the results, findings reveal that several low-lying areas are likely to experience substantial impact as the extent of flooding increases more significantly over time. These results show the urgency of developing adaptive strategies to mitigate the long-term impacts of climate change on Long Island. Future work should involve refining predictions with more localized data and exploring potential mitigation measures to protect vulnerable regions.

## Introduction

- Over the next 25 years, sea levels around Long Island are expected to rise 13 to 25 inches, and as much as 69 inches by the end of the century.
- Earth's temperature has risen by an average of 0.11° Fahrenheit (0.06° Celsius) per decade since 1850, or about 2° F in total. The rate of warming since 1982 is more than three times as fast: 0.36° F (0.20° C) per decade.
- Temperature on Long Island is projected to increase between 3.8°F and 5.8°F by the 2050s and between 5.1°F and 9.5°F by the 2080s compared to the 1981–2010 average.

## Methodology

- Global Mapper
  - Creation of path profile along Long Island
  - Digital elevation models of Long Island based on the areas of study using Shuttle Radar Topography Mission data (SRTM; 30m/pixel resolution).
- IPCC AR6 Sea Level Projection Tool
  - Sea level projections associated with the Intergovernmental Panel on Climate Change Sixth Assessment Report
  - Summary of relative sea level change projections of 6 sites from the study area
- NYSDEC
  - Predictions of sea level rise on long island

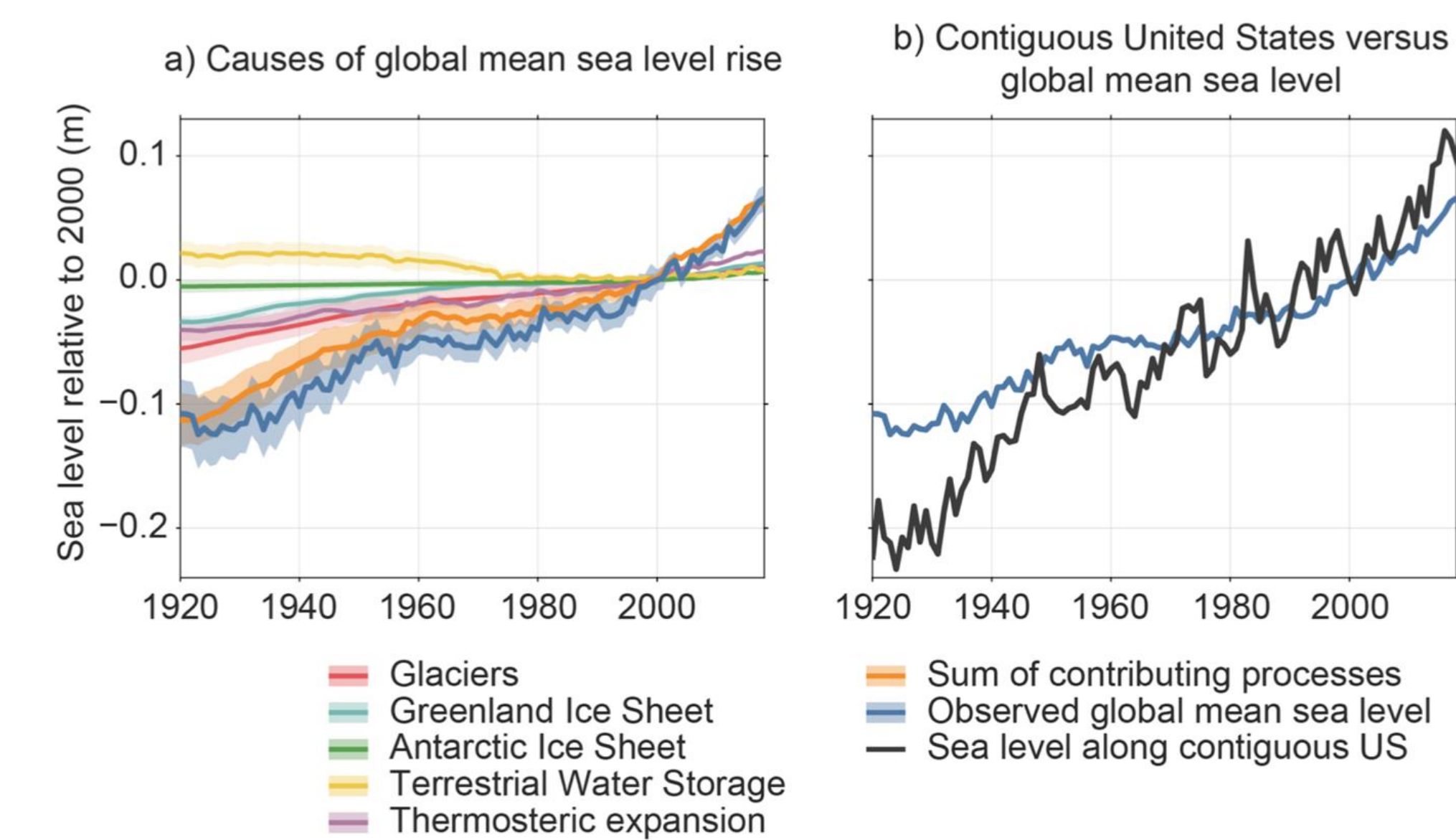


Figure 1. a) Observed annual global mean sea level (GMSL) Change from global tide gauges (blue line), along with the sum (orange line) of contributions from thermal expansion (thermostatic) and four distinct water-mass-driven increases in GMSL. b) GMSL change (blue line) as shown in a) with the annual average relative sea level change measured by tide gauges around the contiguous United States (black line; with a linear regression estimate of 28 cm of sea level rise from 1920 to 2020).

Figure citation (Adaptation of Frederikse et al., 2020).

## Results

- Projections indicate potential sea level rise of 27 inches to 114 inches by the end of the century, depending on global temperature increases
- Five different temperature rise scenarios were analyzed, showing varying impacts on sea level
  - Lower temperature scenarios predict gradual sea level increases of 0.5 to 0.9 meter by 2100
  - Higher temperature scenarios predict more rapid and severe increases of 0.9 to 1.3 meters by 2100
- Data shows correlation between rising temperatures and accelerated sea level changes

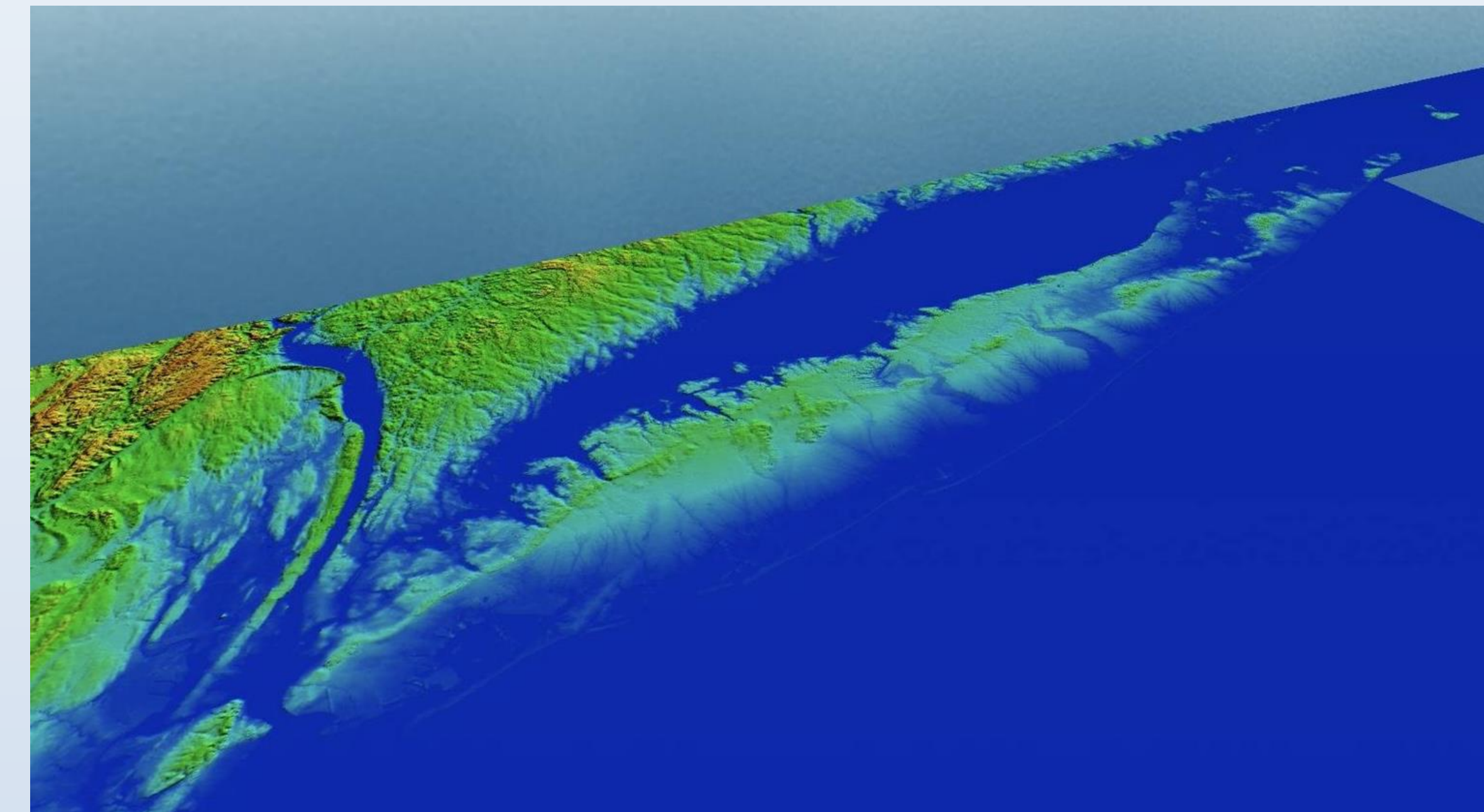


Figure 2. 3D elevation model of area of study (Long Island; this study).

| Predictions for Sea Level Rise on Long Island |                |                       |                   |                        |                 |                           |
|-----------------------------------------------|----------------|-----------------------|-------------------|------------------------|-----------------|---------------------------|
| Time Interval                                 | Low Projection | Low-Medium Projection | Medium Projection | High-Medium Projection | High Projection | Rapid Ice Melt Projection |
| 2030s                                         | 7 inches       | 8 inches              | 10 inches         | 12 inches              | 14 inches       |                           |
| 2050s                                         | 13 inches      | 15 inches             | 18 inches         | 21 inches              | 25 inches       |                           |
| 2080s                                         | 23 inches      | 26 inches             | 32 inches         | 41 inches              | 48 inches       | 83 inches                 |
| 2100                                          | 27 inches      | 32 inches             | 39 inches         | 54 inches              | 69 inches       | 114 inches                |
| 2150                                          | 42 inches      | 50 inches             | 63 inches         | 94 inches              | 185 inches      |                           |

Figure 3. Projection of sea level rise on Long Island in 6 different conditions, for time interval from 2030s to 2150.

Figure Citation (NYSDEC).

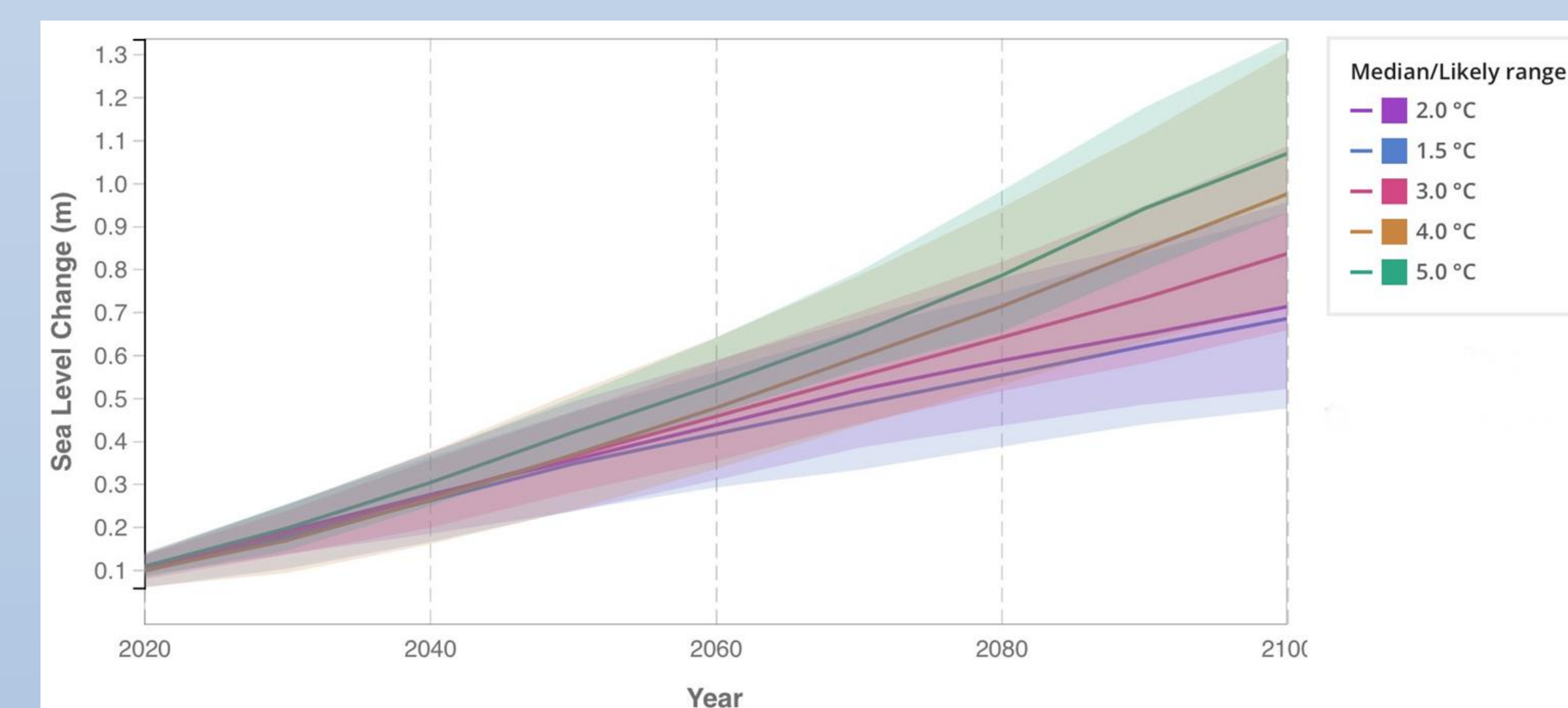


Figure 4. Projection of sea level change in 5 different rise in temperature from 2 to 5 degrees, for time interval 2020 to 2150.

Figure Citation (IPCC AR6 Sea Level Projection Tool).

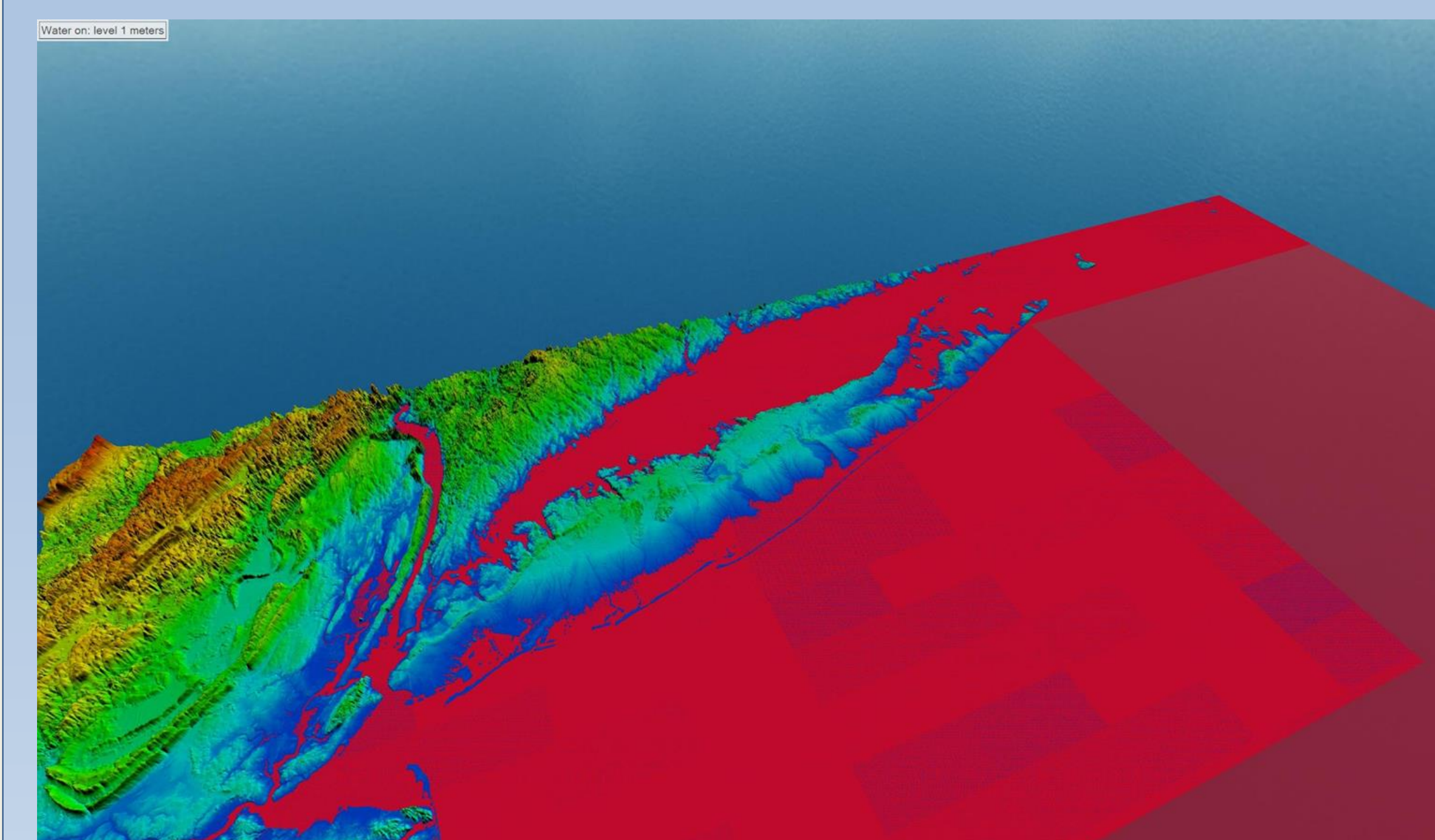


Figure 5. 3D Model of affected area under 1 meter rise in sea level

## Discussion

- Long Island's low lying coastal areas face high risk from future sea level rise, especially in high temperature scenarios
- Projection show significant flooding potential particularly in communities with low elevation
- Projections are limited to current data and assumptions that may evolve, ongoing model updates and localized studies are required

## Conclusions / Significance

- Sea level rise due to climate change threatens Long Island's environment and coastal communities
- This study presents the need for adaptive strategies, such as coastal barriers, to protect lives and properties
- Future studies should refine projections with updated data and explore local adaptation strategies

## Acknowledgements

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## Citations

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